PROJECT REPORT / 20 OCTOBER 2026

Uncertainty-Aware Multi-View Retrieval

English, Hindi and Bengali · Audited public-data CPU pilot

1 Introduction

Search across Indian languages must accommodate script differences, Romanized spellings and language mixing. Exact matching can miss relevant passages; forced rewriting can damage names, numbers or negation. This project preserves original queries, constructs cautious alternatives and combines lexical and neural evidence. Its broader noisy-query motivation is distinct from the original-query pilot evaluated here.

2 Problem statement

Can query-dependent reliability weighting outperform equal and validation-tuned fixed fusion without harming original-query retrieval? The system must support English, Hindi and Bengali, expose uncertainty and retain passage provenance. Public sources did not supply the independently reviewed bilingual challenge required by the original design. The approved original-query pilot therefore excludes unreviewed noisy variants and consistency supervision.

3 Solution approach

Data and isolation. MIRACL queries, relevance judgments and Wikipedia passages [1, 2] form separate monolingual tracks, each with the same fixed 1,500 passages across systems. All selected judged passages, including grade-zero ones, are retained and supplemented with deterministic distractors. Whole query groups preserve their existing assignments; official training supplies the three internal partitions below, and official development supplies the locked final set.

Encoder train	Fusion train	Validation	Final (official dev)
24 groups	16 groups	16 groups*	24 groups

Per language. *Hindi has 17 validation queries in 16 groups; total validation queries = 49. Final evaluation = 72 original queries/groups (24 per language).

Retrieval. Protected views feed four possible branches: original-query BM25, distinct native-view BM25, Roman-view character 3–5-gram TF-IDF and dense E5. Dakshina training lexicons [3] and explicitly algorithmic uroman mappings [6] provide a resource baseline, not a validated IndicXlit model. Auxiliary language ID uses public COMI-LINGUA Hindi–English labels [4] and weak Bengali word labels; it has no Bengali gold test. Unavailable branches are masked and active weights renormalized in reciprocal-rank fusion (rank constant 60). Held-out native-to-Roman word matching is low: 15.68% Hindi and 21.72% Bengali.

Neural adaptation. Multilingual E5-small [5] produces normalized 384-dimensional embeddings. Frozen and adapted models use identical query/passage prefixes and 128-token right truncation of title plus body (encoding batch 8). Each seed (13, 29) receives one SGD epoch on 72 original training queries: learning rate 0.001, training batch 1 and one judged nonrelevant negative per query. There is no paired-variant or consistency loss.

Fusion and delivery. A linear masked-softmax gate learns branch utility from query, language-ID, transliteration and agreement features on fusion training only. Validation selects gate settings and fixed weights. Both seeds are reported; seed 13/full gate is the predeclared demo, without final-set retuning. The app returns ten source passages, view diagnostics and contributions—not generated answers.

4 Experimental results

Table 1. Final language-macro nDCG@10. Mean of the three language means; higher is better. Lexical scores are unchanged across encoders; dashes indicate not run.

Method	Frozen E5	Adapted seed 13	Adapted seed 29
Original-query BM25	0.585748	0.585748	0.585748
Protected Roman character TF-IDF	0.362282	0.362282	0.362282
Original-query dense E5	0.729563	0.729415	0.728356
Equal fusion	0.645935	0.641743	0.643710
Validation-tuned fixed fusion	0.695316	0.700173	0.699647
Full learned gate	—	0.670070	0.672833
Retrained: no uncertainty features	—	0.665776	0.670133
Retrained: no agreement features	—	0.679432	0.685106

Full-gate **MRR@10 = 0.650265 / 0.651802** and **Recall@100 = 0.995370 / 1.000000** for seeds 13 / 29. Serving returns ten hits; ranking replay evaluates recall at 100.

Findings. Adaptation does not improve dense final nDCG in either seed. The full gate beats equal fusion but trails dense-only and tuned fixed fusion. Against dense-only, its query-level nDCG wins/ties/losses are 18/22/32 (seed 13) and 17/19/36 (seed 29). Removing agreement features improves point estimates; removing uncertainty features slightly lowers them. These retrained ablations are descriptive, not proof of significance or a causal explanation.

Availability matters. Native-view BM25 is masked on all 72 final queries: no distinct safe native view is produced. Character retrieval is unavailable on all 24 Bengali queries and one Hindi query; missing character-only runs score zero rather than being dropped. Thus the experiment does not demonstrate a four-active-branch or reliable Bengali normalization benefit.

Engineering verification. All 13 functional modules are integrated. The recorded final suite passes 484 tests under each of two hash seeds (0, 917), with zero skips and warnings treated as errors. Offline wheel build/install and real cached-E5 checks pass. The real-data app passes six query cases, twelve search invocations and 21 HTTP requests across all three languages, including API/HTML, health and invalid-input checks. Independent metric recomputation passes; 729 frozen input files remain intact [7].

5 Conclusion

The project delivers an auditable end-to-end multilingual passage retriever, reproducible training, controlled comparisons and an inspectable search app. The proposed retrieval-quality gains are **not demonstrated** by this pilot: frozen dense E5 is strongest by macro nDCG, and a simple tuned fixed fusion outperforms the full learned gate. Working software and successful tests do not establish research superiority.

Limits and next work. These small, judged-passage-retaining collections are not full MIRACL or the prepared 50,000-passage-per-language experiment. E5 already has MIRACL-training exposure, so its baseline is not zero-shot. Seeds share evaluation queries; exact-duplicate grouping is not a semantic paraphrase audit. No natural noisy/code-mixed robustness, cross-lingual, human-challenge or consistency benefit is established. Larger experiments need more capacity and independently reviewed bilingual data; browser visual review is also outstanding.

REFERENCES ONLY

References and data access

Public dataset links, exact resource versions and execution evidence

[1] MIRACL queries and relevance judgments. Project MIRACL. English, Hindi and Bengali original train/development queries and passage judgments. Dataset card declares Apache-2.0.

https://huggingface.co/datasets/miracl/miracl

Browse/download the pinned data files

`Revision: 5be20db9509754dadad47689368639fcec739c00`

[2] MIRACL passage corpus. Project MIRACL. Wikipedia passage text, titles and IDs for the three monolingual collections. Card declares Apache-2.0; preserve underlying Wikipedia attribution.

https://huggingface.co/datasets/miracl/miracl-corpus

Browse/download the pinned corpus files

`Revision: d921ec7e349ce0d28daf30b2da9da5ee698bef0d`

[3] Dakshina v1.0. Roark et al. (2020), Google Research. Hindi/Bengali native–Roman word lexicons; training pairs supply attested mappings, not retrieval labels. CC-BY-SA-4.0.

https://github.com/google-research-datasets/dakshina

Official download: dakshina_dataset_v1.0.tar
Acquired lexicon members: GCS generation 1590618303711280; member checksums recorded locally, not a verified full-archive MD5.

[4] COMI-LINGUA. LingoIITGN. Human Hindi–English token annotations used for auxiliary language-ID training/calibration/testing; not Bengali retrieval gold. CC-BY-4.0.

https://huggingface.co/datasets/LingoIITGN/COMI-LINGUA

Browse/download the pinned data files

`Revision: 0214cb358e59e36da60e01d3d59ddd24df897749`

[5] Multilingual E5-small. intfloat. Pretrained 384-dimensional retrieval encoder and model card; prior training includes MIRACL. MIT model license.

https://huggingface.co/intfloat/multilingual-e5-small

`Revision: fd1525a9fd15316a2d503bf26ab031a61d056e98`

[6] uroman 1.3.1.1. Ulf Hermjakob, USC Information Sciences Institute. Algorithmic out-of-vocabulary Romanization; bundled USC permissive license and acknowledgment apply.

https://github.com/isi-nlp/uroman

Acknowledgment: This project uses the universal romanizer software 'uroman' written by Ulf Hermjakob, USC Information Sciences Institute (2015-2020).

[7] Local execution and audit artifacts (20 October 2026). Paths relative to the project checkout; these are local evidence, not public dataset URLs.

```
artifacts/study/cpu-pilot/final-report/report.json
artifacts/checks/cpu-pilot-stage6-complete-2026-10-20.json
artifacts/checks/engineering-final-2026-10-20.json
artifacts/checks/cpu-pilot-real-app-2026-10-20.json
artifacts/checks/cpu-pilot-final-independent-audit-2026-10-20.json
```